

optimization objective for the whole system can be written as:

$$\max_{\theta} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \alpha^t R_t^{\Pi^{\theta}} \right] \quad (2)$$

is a risk-sensitive optimization problem that leverages textual gradient descent, fundamentally differing from DRL algorithms designed for POMDPs. Further details on the textual gradient descent approach are provided in the Appendix A.2.

3 Architecture of FINCON

In this section, we present the architecture of FINCON using a two-level hierarchy. First, we describe the hierarchical framework for coordinating the agents' synchronous work and communication. Then, we elaborate on the functionalities of each module that constitutes each agent in FINCON. Finally, we aim to elaborate on how FINCON solves the objective function expressed as Equation (2) through a verbal reinforcement approach.

3.1 Synthesized Multi-agent Hierarchical Structure Design

The agent system of FINCON consists of two main components: the Manager-Analyst Agent Group component and the Risk-Control component.

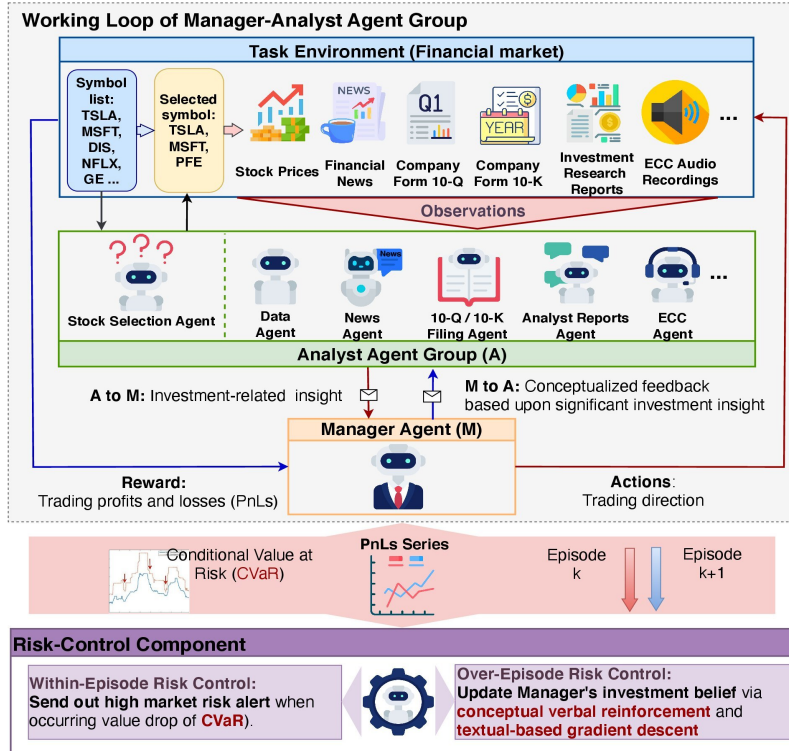


Figure 2: The detailed architecture of FINCON contains two key components: Manager-Analyst agent group and Risk Control. It also presents the between-component interaction of FINCON and decision-making flow.

3.1.1 Manager-Analyst Agent Group

Analogous to human investment firm, FINCON establishes a unique hierarchical structure to organize its multi-agent system, synthesizing their efforts to achieve superior decision-making outcomes.

The primary goal is to enhance information presentation and comprehension while minimizing unnecessary communication costs. The working mechanism of each agent is illustrated in Figure 2.

Analyst Agents. In FINCON, analyst agents distill concise investment insights from large volumes of multi-source market data, each focused on a specific trading target. To ensure high-quality reasoning by reducing task load and sharpening focus, each agent processes information from a single source in a uni-modal fashion, providing pre-specified outputs based on prompts. This setup mimics an efficient human team, where each analyst specializes in a specific function, filtering out market noise and extracting key insights. These agents assist the manager agent by consolidating denoised investment information from multiple perspectives. We implement seven distinct types of analyst agents using LLMs, each producing unique investment insights, as shown in the upper section of Figure 2. Based on input modalities, three textual data processing agents extract insights and sentiments from daily news and financial reports. An audio agent uses the Whisper API to interpret investment signals from earnings call recordings. Additionally, a data analysis agent and a stock selection agent compute critical financial metrics, such as momentum and CVaR, using tabular time series data. The stock selection agent also oversees portfolio selection by applying the classic risk diversification method in quantitative finance [1].

Manager Agent. In FINCON, the manager agent acts as the sole decision-maker, responsible for generating trading actions for sequential financial tasks. For portfolio management, it calculates portfolio weights using convex optimization techniques constrained by directional trading decisions (see optimization problem as presented in Formula (1)). Four key mechanisms support each decision: 1) Consolidating distilled insights from multiple analyst agents. 2) Receiving timely risk alerts and conceptual investment updates from the risk control component. 3) Refining its investment beliefs about the influence of different information sources on trading decisions for specific targets. 4) Conducting self-reflection by reviewing reasoning outcomes from previous trading actions.

3.1.2 Risk-Control Component

We have innovatively designed a dual-level risk-control mechanism consisting of within-episode and over-episode risk management. The within-episode mechanism detects market risk within a single training episode, allowing the manager agent to promptly adjust trading actions to mitigate potential losses by accounting for short-term trading performance and market fluctuations. This mechanism also operates during the testing phase. In contrast, the over-episode mechanism functions exclusively during the training stage, providing prompt optimization guidance by comparing the trading performance of the current episode with the previous one. This reflection enables the manager agent to update its investment beliefs based on performance differences. By drawing on prior observations of market risk and profitability patterns, these two mechanisms help avoid repeated investment errors, thereby enhancing future returns.

Within-Episode Risk Control: The within-episode risk alert is triggered by a sudden drop in the CVaR value. Conditional Value at Risk (**CVaR**) represents the average of the worst-performing 1% of daily trading Profits and Losses (**PnLs**). A decrease in CVaR typically indicates that recent trading decisions have led to PnLs within this bottom percentile, signaling a potentially high-risk market condition. When this occurs, the manager agent adopts a risk-averse stance for that day’s trading actions, regardless of the prior risk status.

Over-Episode Risk Control: The over-episode investment belief updates facilitate adjustments in the emphasis placed on analysts’ information distillation and the manager’s action generation. Through the **Actor-Critic** mechanism, FINCON episodically optimizes its investment strategy for a given trading target, as defined by objective (Equation (2)), by reflecting on a series of winning and losing actions. This episodic reflection is powered by a unique **Conceptual Verbal Reinforcement (CVRF)**. CVRF assesses the performance of consecutive training episodes by analyzing the information perspectives provided by analysts and reflected in the manager’s decision-making. It then conceptualizes and attributes the evaluation outcomes to these specific aspects. By comparing the conceptualized insights from more profitable versus less profitable episodes, the system informs both the manager and analyst agents about necessary belief adjustments, helping prioritize the most relevant market information for increased profitability, as detailed in Algorithm 1. CVRF leverages text-based gradient descent to offer optimal conceptual investment guidance for the manager agent, refining prompts with the latest investment beliefs. The guidance is organized according to perspectives provided by

the respective analyst agents, key financial indicators (such as historical momentum), or other crucial viewpoints.

Factor	Gradient-based model optimizer	LLM-based prompt optimizer
Upgrade direction	Model value gradient momentum	Prompt reflection trajectory
Update method	Learning rate descent	Overlapping percentage of trading decisions

Table 1: Analogy between glossaries in model optimizer and prompt optimizer.

These belief updates are first received by the manager agent and then selectively propagated to relevant agents, minimizing over-communication. Unlike the text-based gradient descent proposed by Tang et al.[28], which uses prompt editing distance as a learning rate, we derive investment belief updates by measuring the overlapping percentage of trading actions between two consecutive training trajectories at each belief update, as presented in Table 1. This approach has proven effective in improving the performance of a synthesized agent system, where each worker has a clearly defined and specialized role. The above describes the workflow of FINCON during the training stage, while the workflow during the testing stage is detailed in the Appendix A.3.

Algorithm 1 Training Stage Algorithm of FINCON: Conceptual Verbal Reinforcement using Textual-based Gradient Descent

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Initialize manager-analysts component  $\{M_{pr}^i\}_{i=1}^I$  &  $M_a$ , and risk-control component  $M_r$ .
Initialize trading start date  $s$ , stock pool of portfolio and portfolio weights  $w_0 = \mathbf{0}$ .
Initialize Prompts  $\theta$ , policy  $\Pi_\theta$ .
while episode  $k < Max$  do
  for  $0 \leq t \leq T$  do
    Run policy  $\Pi_\theta$  (collecting daily PnL  $r_t$ , portfolio weights  $w_t$  and daily CVaR value  $\rho_t$ ).
    if  $\rho_t < \rho_{t-1}$  or  $r_t < 0$  then
      Trigger  $M_a$  self-reflection and generate self-reflection text  $B_t$ .
    end if
    Get the investment trajectory  $\mathcal{H}_k$  and calculate the objective function value (Function (2)).
  end for
  Compare the objective function values of episodes  $k - 1$  &  $k$ , and decide which episode has higher performance;
  Pass sustained profitable and losing trades from two episodes  $\mathcal{H}_{k-1}$  &  $\mathcal{H}_k$  into risk-control component  $M_r$ ;
  Guide  $M_r$  to summarize conceptualized investment insights  $\{c_{k-1}^1, \dots, c_{k-1}^n\}$  &  $\{c_k^1, \dots, c_k^m\}$ ;
  Compare two sets of conceptualized insights and give the reasoning for higher performance (providing textual optimization direction, i.e. meta prompt);
  Calculate the overlapping percentage between trading decision sequences from two episodes (providing the learning rate  $\tau$ );
  Update the prompts by textual gradient-descent:
     $\theta \leftarrow M_r(\theta, \tau, meta\ prompt)$ .
end while

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3.2 Modular Design of FINCON Agents

Here, we explain the modular design of FINCON agents. Inspired by the recent works of Park et al. [44] and Sumers et al. [45] on developing the cognitive structure of language agents for human-like behavior, agents in FINCON integrate four modules to support their necessary functionalities, along with a shared general configuration, as detailed in Appendix A.4:

General Configuration and Profiling Module. This module defines task types (e.g., stock trading, portfolio management) and specifies trading targets, including sector and performance details. The profiling module outlines each agent’s roles and responsibilities. The concatenated textual content from these parts is used to query investment-related events from the agents’ memory databases.

Perception Module. This module defines how each agent interacts with the market, specifying the information they perceive, receive, and communicate, with interactions tailored to each agent’s role. In detail, it converts raw market data, feedback from other agents, and information retrieved from the memory module into formats compatible with large language models, enabling them to process these inputs effectively.

Memory Module. The memory module comprises three key components: